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C PROGRAMMING

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QUESTIONS 18 / 18 completed

Natural

NATURAL

Write a program that prints the first n natural numbers using a for loop.

PROGRAM EDITOR

Run | Save

```c
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter n: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive number\n");
    } else {
        printf("First %d natural numbers:\n", n);
        for (i = 1; i <= n; i++) {
            printf("%d ", i);
        }
    }

    return 0;
}
```

INPUT

6

OUTPUT

```
Enter n: First 6 natural numbers:
1 2 3 4 5 6
```

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QUESTIONS 18 / 18 completed

Sum

SUM

Write a program that calculates the sum of the first n natural numbers using a while loop.

PROGRAM EDITOR

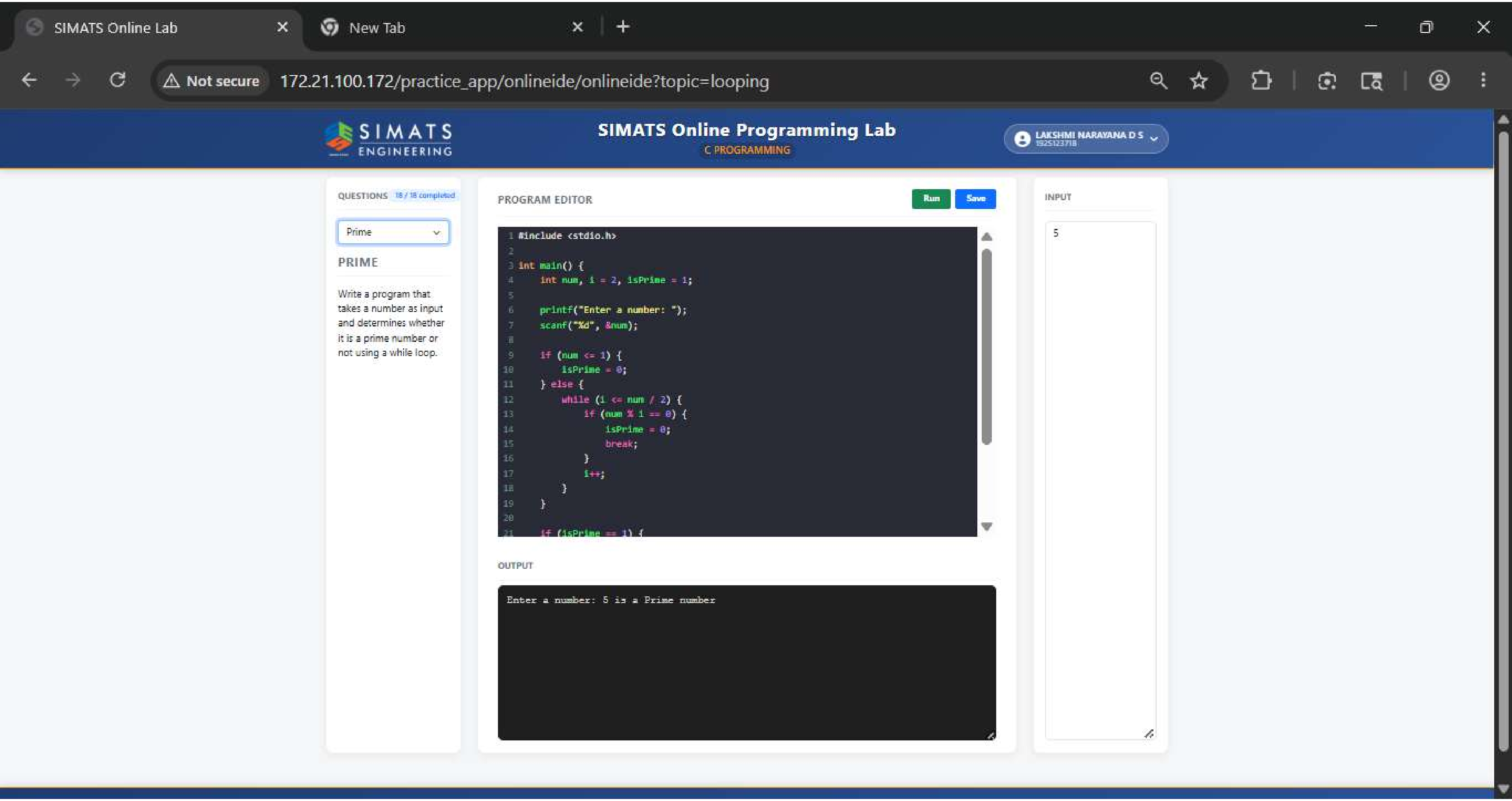
Run Save

```
#include<stdio.h>
int main()
{
  int n,i,sum=0;
  printf("enter the number of values:\n");
  scanf("%d",&n);

  while(i<=n)
  {
    printf("%d",i);

  }
}
```

OUTPUT

INPUT

Your INPUT goes here!



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QUESTIONS
18 / 18 completed
Prime
PRIME
Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.
PROGRAM EDITOR
Run
Save
#include <stdio.h>
int main() {
int num, i = 2, isPrime = 1;
printf("Enter a number: ");
scanf("%d", &num);
if (num <= 1) {
isPrime = 0;
} else {
while (i <= num / 2) {
if (num % i == 0) {
isPrime = 0;
break;
}
i++;
}
}
if (isPrime == 1) {
OUTPUT
Enter a number: 5 is a Prime number
INPUT
5

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QUESTIONS 18 / 18 completed

Count Vows

COUNT VOWS

Write a program that takes a string as input and counts the number of vowels in the string using a for loop.

PROGRAM EDITOR

Run Save

```c
#include <stdio.h>

int main() {
    char str[100];
    int i, count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
            str[i] == 'o' || str[i] == 'u' ||
            str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
            str[i] == 'O' || str[i] == 'U') {
            count++;
        }
    }

    printf("Number of vowels = %d\n", count);

    return 0;
```

OUTPUT

```
Security Error: Dangerous code pattern detected (#gets\s*\(#i).
```

INPUT

A

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QUESTIONS 18 / 18 completed

Fibonacci

FIBONACCI

Write a program that prints the Fibonacci series up to a given number using a while loop.

PROGRAM EDITOR

Run Save

```c
#include<stdio.h>
int main()
{
  int n,a,b=1,c;
  printf("enter the number of values:\n");
  scanf("%d",&n);

  while(a<=n)
  {
    printf("%d\n",a);
    c=a+b;
    a=b;
    b=c;
  }
}
```

OUTPUT

```
enter the number of values:
0
1
1
2
3
5
8
13
21
```

INPUT

999

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QUESTIONS 18 / 18 completed

Diamond

DIAMOND

Write a program that prints a diamond pattern of stars using nested loops.

PROGRAM EDITOR

Run Save

```c
#include <stdio.h>

int main() {
    int n, i, j;

    printf("Enter number of rows:\n");
    scanf("%d", &n);

    for(i = 1; i <= 2 * n - 1; i++) {
        int stars;

        if(i <= n)
            stars = i;
        else
            stars = 2 * n - i;

        for(j = 1; j <= n - stars; j++)
            printf(" ");

        for(j = 1; j <= 2 * stars - 1; j++)
            printf("*");
```

OUTPUT

```
Enter number of rows:
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

INPUT

5

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QUESTIONS 18 / 18 completed

R- triangle

R- TRIANGLE

Write a program that prints a right-angled triangle pattern of stars using nested loops.

PROGRAM EDITOR

Run Save

```c
#include <stdio.h>

int main() {
    int n, i, j;

    printf("Enter number of rows:\n");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        for(j = 1; j <= i; j++) {
            printf("* ");
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT

```
Enter number of rows:
*
* *
* * *
* * * *
* * * * *
```

INPUT

5

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QUESTIONS 18 / 18 completed

Hollow Diamond

HOLLOW DIAMOND

Write a program that prints a hollow diamond pattern of stars using nested loops.

PROGRAM EDITOR

Run Save

```c
#include <stdio.h>

int main() {
    int n, i, j, space;

    printf("Enter number of rows:\n");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        for(space = 1; space <= n - i; space++)
            printf(" ");

        for(j = 1; j <= 2 * i - 1; j++) {
            if(j == 1 || j == 2 * i - 1)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }

```

OUTPUT

```
Enter number of rows:
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```

INPUT

5

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QUESTIONS 18 / 18 completed

Pascal's triangle

PASCAL'S TRIANGLE

Write a program that prints a Pascal's triangle pattern using nested loops.

PROGRAM EDITOR

Run Save

```c
#include <stdio.h>

int main() {
    int n, i, j, num;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        num = 1;

        // spaces for formatting
        for (j = 0; j < n - i - 1; j++) {
            printf("  ");
        }

        // print numbers
        for (j = 0; j <= i; j++) {
            printf("%4d", num);
            num = num * (i - j) / (j + 1);
        }
```

OUTPUT

```
Enter number of rows:           1
         1   1
       1   2   1
     1   3   3   1
   1   4   6   4   1
```

INPUT

5

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QUESTIONS 18 / 18 completed

zigzag

ZIGZAG

Write a program that prints a zigzag pattern of stars using nested loops.

PROGRAM EDITOR

Run Save

```c
#include <stdio.h>

int main() {
    int i, j, n = 9;  // number of columns

    for (i = 1; i <= 3; i++) {          // 3 rows for zigzag
        for (j = 1; j <= n; j++) {
            if ((i == 1 && j % 4 == 3) ||
                (i == 2 && j % 2 == 0) ||
                (i == 3 && j % 4 == 1)) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT

```
  *   *
 * * * *
*   *   *
```

INPUT

5

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QUESTIONS 18 / 18 completed

Spiral pattern

SPIRAL PATTERN

Write a program that prints a spiral pattern of numbers using nested loops.

PROGRAM EDITOR

Run Save

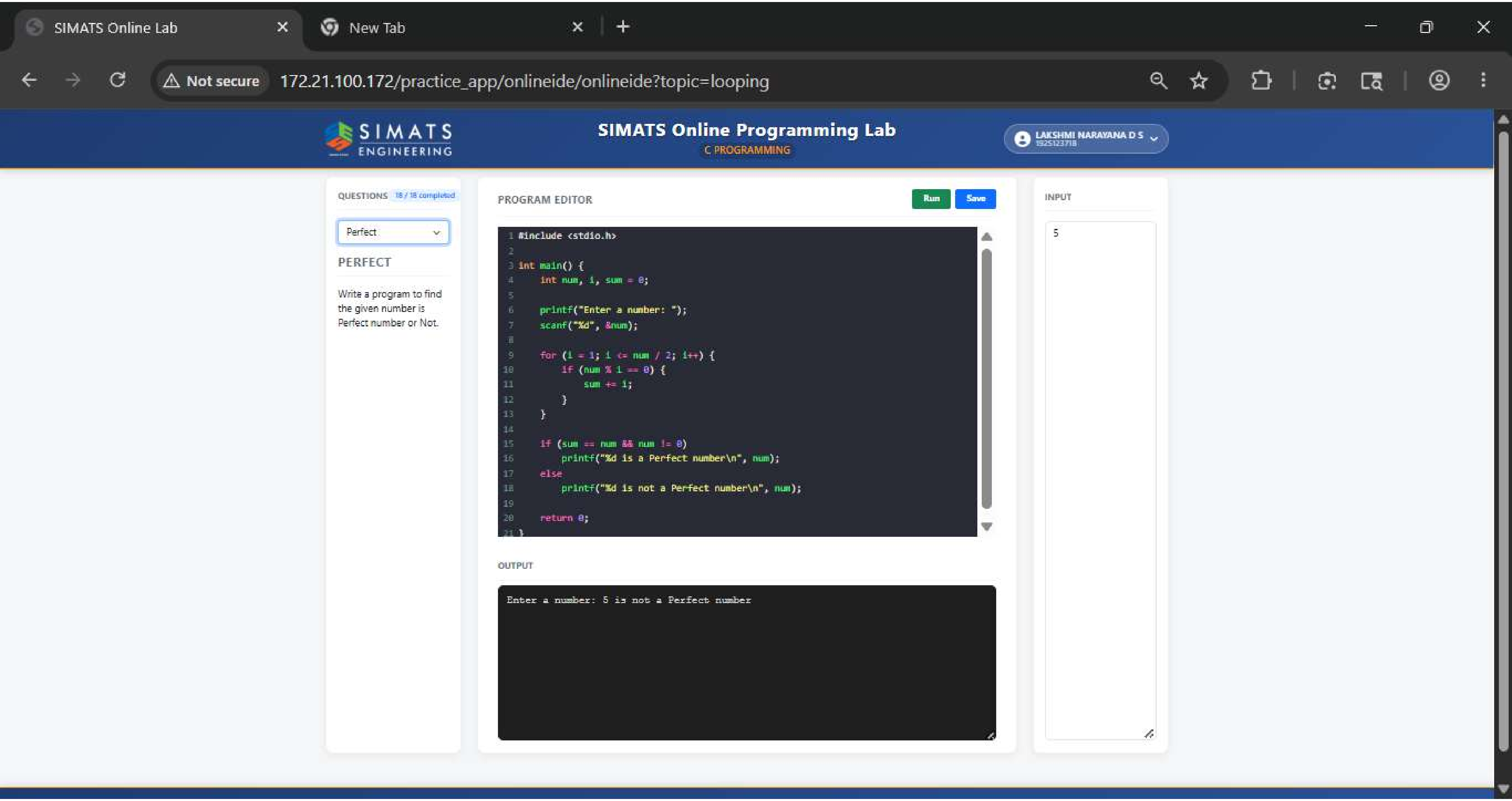
```
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter size (n x n):\n ");
    scanf("%d", &n);

    int top = 0, bottom = n - 1;
    int left = 0, right = n - 1;
    int num = 1;
    int a[n][n];

    while (top <= bottom && left <= right) {

        // left → right
        for (i = left; i <= right; i++)
            a[top][i] = num++;
        top++;

        // top → bottom
```

OUTPUT

```
Enter size (n x n):
   1   2   3   4   5   6
  20  21  22  23  24   7
  19  32  33  34  25   8
  18  31  36  35  26   9
  17  30  29  28  27  10
  16  15  14  13  12  11
```

INPUT

6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QUESTIONS 18 / 18 completed
Perfect
PERFECT
Write a program to find the given number is Perfect number or Not.
PROGRAM EDITOR
Run
Save
#include <stdio.h>
int main() {
int num, i, sum = 0;
printf("Enter a number: ");
scanf("%d", &num);
for (i = 1; i <= num / 2; i++) {
if (num % i == 0) {
sum += i;
}
}
if (sum == num && num != 0)
printf("%d is a Perfect number\n", num);
else
printf("%d is not a Perfect number\n", num);
return 0;
}
OUTPUT
Enter a number: 5 is not a Perfect number
INPUT
5